Microgravity Flight Course - Design, Build, Fly

Syllabus for Graduate Course 16.88J/MAS.868 (MIT), AERO E4450 (Columbia)

Instructors: Professor Cody Paige (Columbia), Professor Dava Newman (MIT)

Class period: Thursdays 1-2:30 pm (MIT, 33-218; Columbia TBD)

Lab/Studio: Thur. 2:30p - 3:40p (Columbia), Tuesdays 1-2:30 pm (MIT)

Course website: On Canvas @MIT; Columbia TBD

Contact: dnewman@mit.edu, c.paige@columbia.edu

Description: Rapid prototyping and engineering skills for experiments operating in altered gravity environments. Logistics, training, and safety pre-approval steps required to meet flight readiness requirements and pass a Technical Readiness Review (TRR). Topics include parabolic flight mechanics; human physiology in reduced gravity; the space environment; mechanical and electrical design for flight hardware; data acquisition and experiment operations; safety analysis and fault tree methods; payload integration requirements. Team-based development of a flight-ready microgravity experiment concept, culminating in a complete Payload Integration Package (PIP) and Critical Design Review (CDR). Guest lectures from aerospace industry and spaceflight professionals. Where funding and flight opportunities permit, student teams may have the opportunity to fly their experiments on an actual microgravity flight the following semester.

At MIT, we will meet on Tuesday and Thursdays (1-2:30 pm in 33-218). Lectures are on Thursdays to provide background knowledge and relevant real-world examples (lectures, readings, videos, and advice from experienced astronauts, explorers, and space mission principal investigators (PIs)). Tuesday will be a 'lab' day to support prototyping and project development (mostly in the second half of the semester) and Thursday is a 'lecture and discussion' day.

Learning Outcomes

- Team design and fabrication of flight-ready hardware for parabolic flight that meets the payload provider (NovaSpace, ZeroG, NASA, etc.) safety and integration requirements.
- Flight experiment application of fault analysis methods to identify and mitigate hazards.
- End-to-end experiment development from concept through Payload Integration Package (PIP) and [future flight Technical Readiness Review].
- Design, Build, Fly (DBF skills) – Hands-on prototyping, ground testing, and verification of microgravity experiment hardware.
- Communication of technical work through formal design reviews (PDR, CDR) and public-facing communications on experiment/payload descriptions.

This class is not intended to teach the fundamentals of aerospace design, embedded programming & circuits, and rapid prototyping. On most topics, we will assume prior knowledge and skills. If you are new to these skills, we recommend taking "How to Make (Almost) Anything" as a prerequisite.

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https://ocw.mit.edu/courses/mas-863-how-to-make-almost-anything-fall-2002/pages/syllabus/

https://fab.cba.mit.edu/classes/MAS.863/

Motivation: The commercial space industry is growing rapidly, and there is strong demand for engineers who understand how to design hardware and experiments for microgravity. Parabolic flights, analog space environments, and the International Space Station (ISS) all offer accessible routes for university-level research, yet few programs prepare students with the end-to-end skills needed to take an idea from concept to flight readiness.

This course bridges that gap by:

- Providing the technical foundations of microgravity experiment design, from structural and electrical engineering to safety qualification and operations, through lectures, guest experts, and hands-on design periods.
- Tasking student teams with designing and building a prototype experiment that meets real flight provider requirements, culminating in a Critical Design Review (CDR) judged against industry standards.

Grading: The learning approach to be taken in this course is to learn by doing and from others. Grading is based on individual assignments, team-based design reviews, and a final experiment design package. Homework assignments will be to ensure students have the needed knowledge for prototyping, including CAD and electronics design, as well as to ensure experiment design progression through an ideation assignment and a prototype test-run assignment. Most labs will operate like an architecture studio where students will be able to work on their projects, access shop/maker spaces, and the professor and advisors will circulate for student updates answering questions and providing feedback. Occasional labs will have focused training or project goals (specified below).

Course Requirements & Evaluation	**Grade Percentage**
Attendance and participation	10%
Homework assignments / Lab Write-Ups	20%
Preliminary Design Review (PDR) — team presentation	20%
Critical Design Review (CDR) — team presentation and prototype	35%
Final Payload Integration Package (PIP) documentation	15%

Texts: Papers will be assigned on a weekly basis by the student groups in topics relating to their proposed experimental concepts, as well as preparatory reading assigned for guest lecturers.

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Course Syllabus @MIT in 33-218

Thursdays will be lectures. Tuesdays are lab/studio days @MIT.

Week	Title	Lecturers	Topics
[1] **10 Sept.**	**Introduction to Microgravity**	**Course Instructors** **Manuel Castro**	• Overview of the course and project expectations • History of parabolic flight and microgravity research • Parabolic flight opportunities: trajectories, quality of microgravity, aircraft types • M. Castro: Parabolic flight leading to and ISS experiment
[2] **15 Sept. Tuesday** **17 Sept. Thursday**	**Experiment Concept Development**	**Course Instructors** **Professors Dava Newman Cody Paige**	• Overview of ISS research platforms and commercial spaceflight opportunities • Literature review strategies for space experiments • Introduction to the design challenge: propose a microgravity experiment Science question formulation and hypothesis development • Defining success criteria and measurable outcomes • Workshop: team concept pitches and peer critique • **Assignment:** 2-page experiment concept **due 9/18/26**
	Opportunity	**24-25 Sept.**	**Boston Human Spaceflight Symposium** https://bostonspaceflight.org
[3] **24 Sept. Thursday**	**Engineering and Human Performance in Reduced Gravity**	**Andrew Abercrombie** **Course Instructors**	• Case studies from NASA and ESA parabolic flight campaigns • Analog Environments (underwater, desert, arctic) • Introduction to Payload User's Guide (PUG) requirements • Discussion: Justification for flying in Reduced Gravity
[4] **1 Oct.**	**Case study: MIT *SkinSuit*** **Physiology, Medicine, & Mechanical Design**	**Rachel Bellisle** **Course Instructors**	• Human physiology in microgravity: musculoskeletal, cardiovascular, vestibular, SANS • Structural loads, vibration, acceleration flight env. • Factor of safety requirements and structural calculations • Material selection for flight hardware • Parabolic flight campaigns • ISS AX-2 flight
[5] **6th Oct** **8th Oct**	**Sensing, Actuation, and Electronics**	**Course Instructors**	• **10/6/26 Lab: hands-on CAD/Gen design session** • 10/8/26 Sensor selection for microgravity experiments (IMU, force, imaging) • Embedded systems and microcontrollers (Arduino, Raspberry Pi, STM32) • Power budgeting and battery management for flight hardware • Best practices for wiring, connector selection, and EMI mitigation • **Lab Write-Up: CAD/Gen Design concepts and down select**
[6] **Oct 15th**	**Safety, Fault Analysis, and Regulatory Compliance**	**Course Instructors**	• Parabolic flight safety requirements and hazard categories • Fault tree analysis and failure mode and effects analysis (FMEA) • Pressure vessel, fluid containment, and chemical hazard rules

			• Electrical safety: fusing, ground faults, arcing prevention • Completing the Payload Integration Package (PIP): section-by-section walkthrough
[7] **20th Oct** **22nd Oct**	**Rapid Prototyping Methods**	**Course Instructors**	• **10/20/26b Labs: circuit prototyping session, prototype build session** • 3D printing: FDM vs. SLA, design for print, tolerancing • CNC machining and sheet metal for flight structures • Bonding, fastening, and surface finishing for flight hardware
[8] **27th Oct** **29th Oct**	**Preliminary Design Review (PDR)**	**Course Instructors** **Invited Reviewers**	• **10/27/26 DBF Open Lab** • In-class PDR presentations: team presentation + Q&A • Review criteria: scientific merit, feasibility, safety compliance, completeness • Feedback and iteration planning • Guest reviewers: faculty. industry or NASA flight ops experts (TBD)
[9] **3rd Nov** **5th Nov**	**PDR Debrief** **Data Acquisition and Experiment Control**	**Course Instructors**	• **11/3/26 DBF Open Lab** • Real-time data logging: timing, synchronization, storage • Camera systems and high-speed imaging for microgravity • Remote monitoring and onboard autonomy vs. manual operation • Software tools: LabVIEW, Python, MATLAB for experiment control
[10] **10th Nov** **12th Nov**	**Testing and Verification**	**Course Instructors**	• **11/10/26 DBF Testing Lab** • Ground controls • Environmental testing: vibration, thermal, EMI • Test-as-you-fly philosophy and acceptance testing • Documenting test results for the Technical Readiness Review (TRR)
[11] **19th Nov**	**Cosmos** **Human Factors and Operations**	**Guest Lecturers**	• Flyer roles and responsibilities on a parabolic flight • Crew interfaces: displays, controls, and procedures • Stowage, restraint, and emergency procedures • Operational procedures documentation • **Guest lecture**: parabolic flight veterans (TBD) ***[Aurelia and MIT Novaspace Parabolic Flights]***
[12] **24th Nov**	**Space Experiment Heritage and Future Comm. Platforms**	**Course Instructors**	• Shuttle, MIT, ISS flight experiments (development and results) • Commercial destinations and new launch opportunities • Guest lecture: Commercial spaceflight representative (TBD)
[13] **Dec 1st** **Dec 3rd**	**Critical Design Review (CDR)**	**Course Instructors** **Invited Reviewers**	• In-class CDR presentations: team presentations + Q&A • Review criteria: prototype completeness, safety package, PIP draft, operations readiness • External reviewers from academia, aerospace industry (TBD) • Post-CDR debrief: common lessons learned • Path forward: finalizing PIPs, scheduling flights, post-flight data analysis • Course retrospective and student reflections • **Final PIP documentation due (10 Dec 2026**)

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IAP/S'27	***DBF FRR***		• Flight selection based on EOT CDRs, Flight Readiness Review, Funding

Readings:
Week 1. Reading: Syllabus. Karmali and Shelhamer Parabolic Flight Dynamics (Acta Astra, 2008) Lab: None this week

Week 2.
Lab 09/15/26: Experiment Concept Development.
Readings: Newman, Amir, Beck, Astronaut Induced Loads in Microgravity, J Spacecraft & Rockets, 2001.
Amir and Newman, Astronaut Motion Review, Acta Astronautica, 2000.

09/17/26 Readings: M. Castro Concept Paper. Parabolic flight provider's "Payload User Guide (PUG)"

Week 3. Reading: A. Abercomby

Week 4. Reading: R. Bellisle

Week 5. Readings: HSL flight experiments.

Additional Readings:

HSL historical pubs / Prof. Dava Newman pubs

MOXIE – Mars ISRU flight experiment

Operating Manual for Spaceship Earth – R. Buckminster Fuller

Cosmos